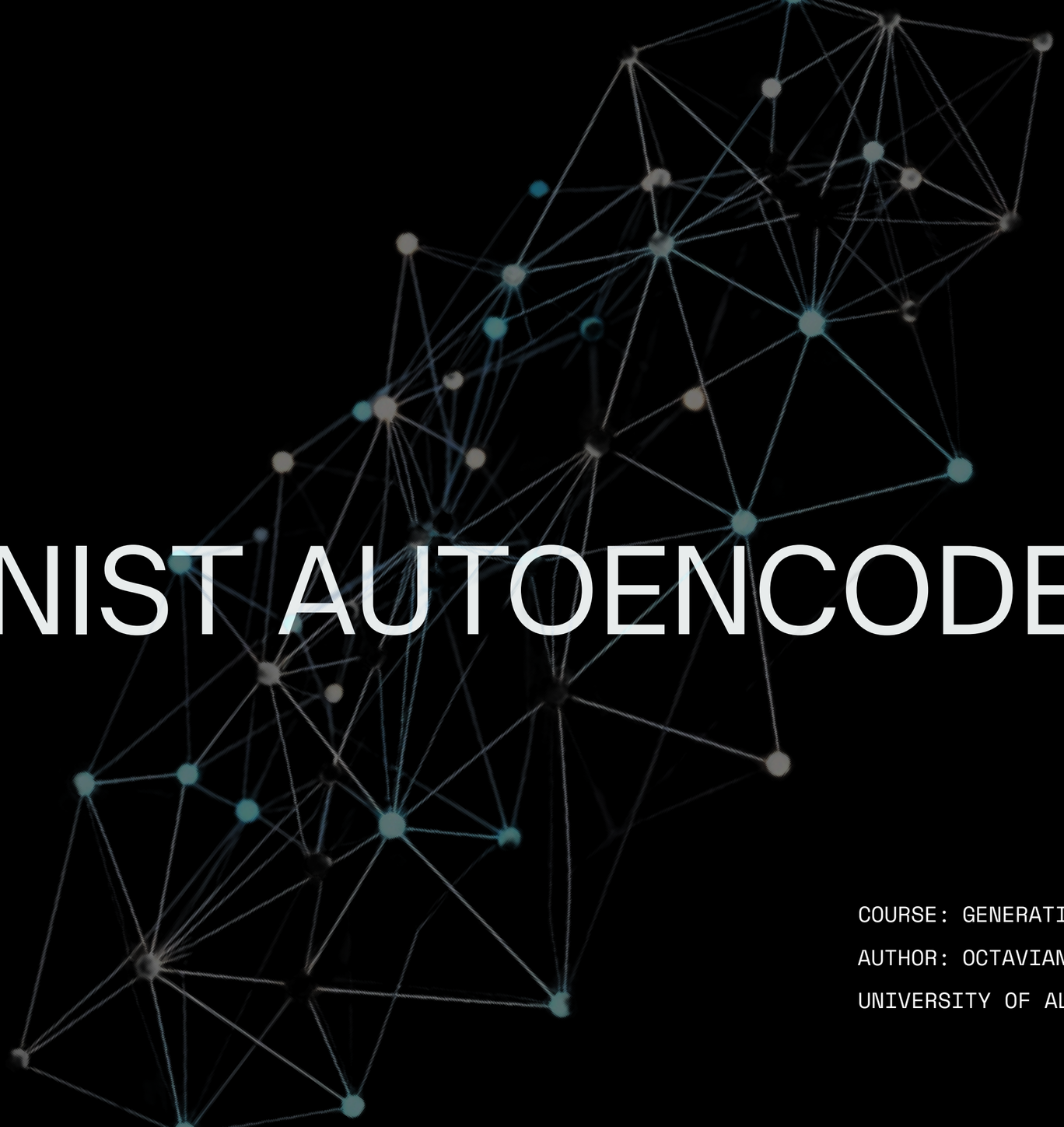


MNIST AUTOENCODER

COURSE: GENERATIVE ARTIFICIAL INTELLIGENCE

AUTHOR: OCTAVIAN POPUIAC — 79911

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HOW DOES AN AUTOENCODER WORK?

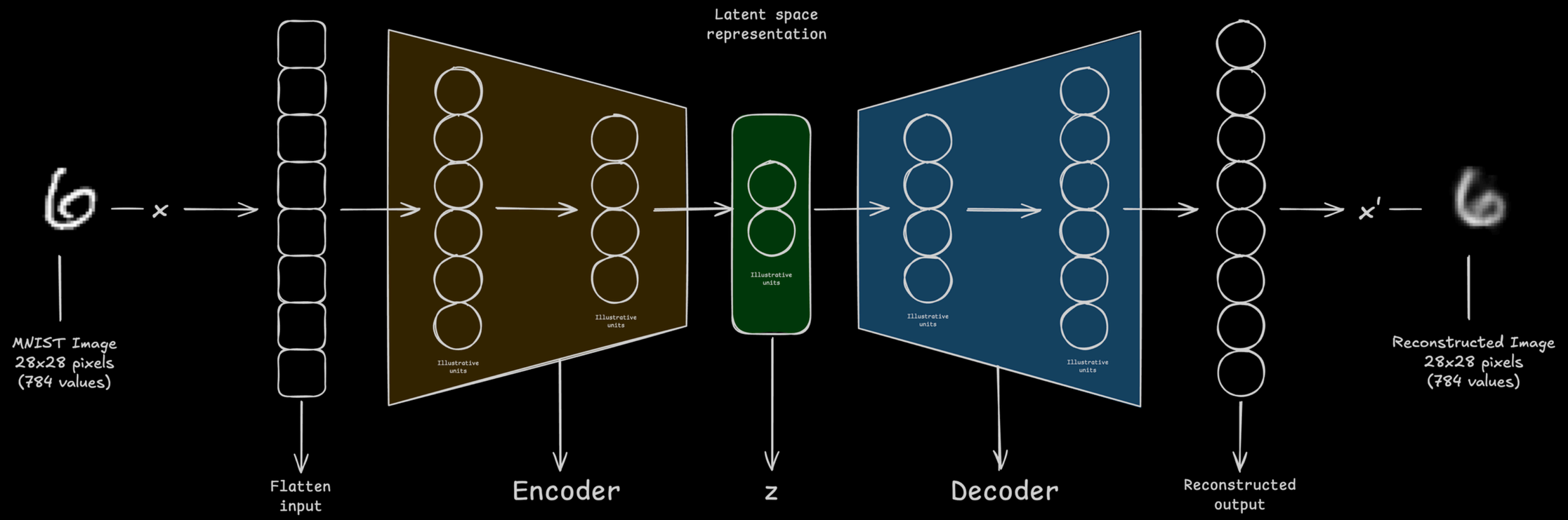


Figure 1 - Illustrative autoencoder architecture with a two-dimensional bottleneck.

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EXPERIMENTAL SETUP

Dataset → 50k train / 10k val / 10k test

Architectures → 1L / 2L / 3L

Regularization → None / L1 / L2

Seeds → 42, 123, 2026

9 configs x 3 seeds = 27 runs

In/Out shape	28x28x1
Pixel normalization	[0,1]
Latent dimension	2
Hidden/Latent activation	ReLU
Out activation	Sigmoid
Reconstruction loss	MSE
Batch size	128
Max epochs	30
Early stopping	Validation MSE - patience: 5
Optimizer	Adam - 0.001

Table 1 - Model and training settings.

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COMPARED ARCHITECTURES

Depth	Encoder	Decoder
1	784 → 128 → 2	2 → 128 → 784
2	784 → 256 → 128 → 2	2 → 128 → 256 → 784
3	784 → 256 → 128 → 64 → 2	2 → 64 → 128 → 256 → 784

Table 2 - Encoder and decoder architectures compared in the experiments. (Depth excludes the lattent layer)

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EFFECT OF DEPTH AND REGULARIZATION

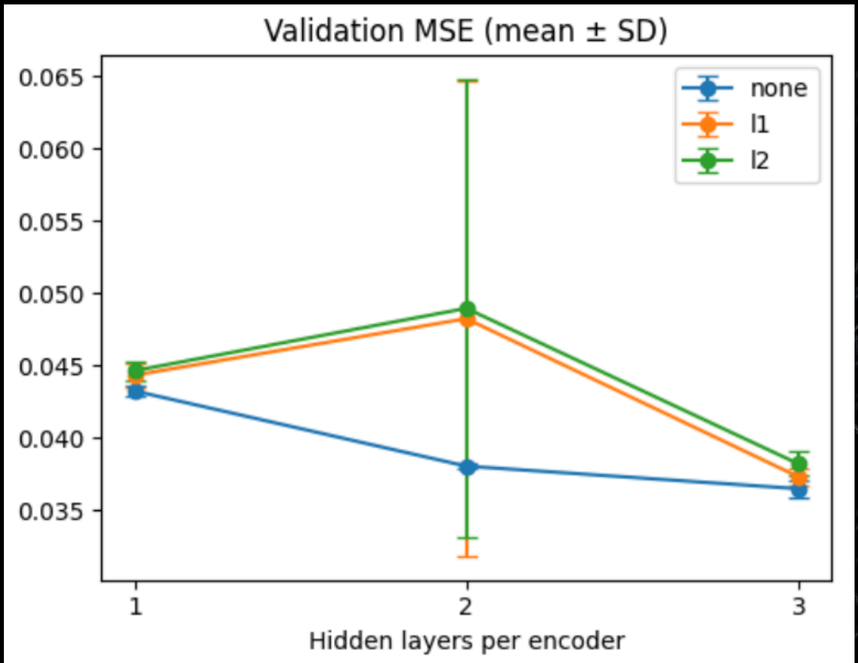


Figure 2 - Validation MSE by encoder depth and regularization: mean ± sample standard deviation across three seeds.

Depth improves the unregularized baseline
Validation MSE decreases from 0.04326 to 0.03650

Best tested configuration
3 hidden layers, no regularization

Regularized depth-2 models show high variability
Large standard deviations are driven by the stalled runs with seed 2026

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TRAINING STABILITY ACROSS SEEDS

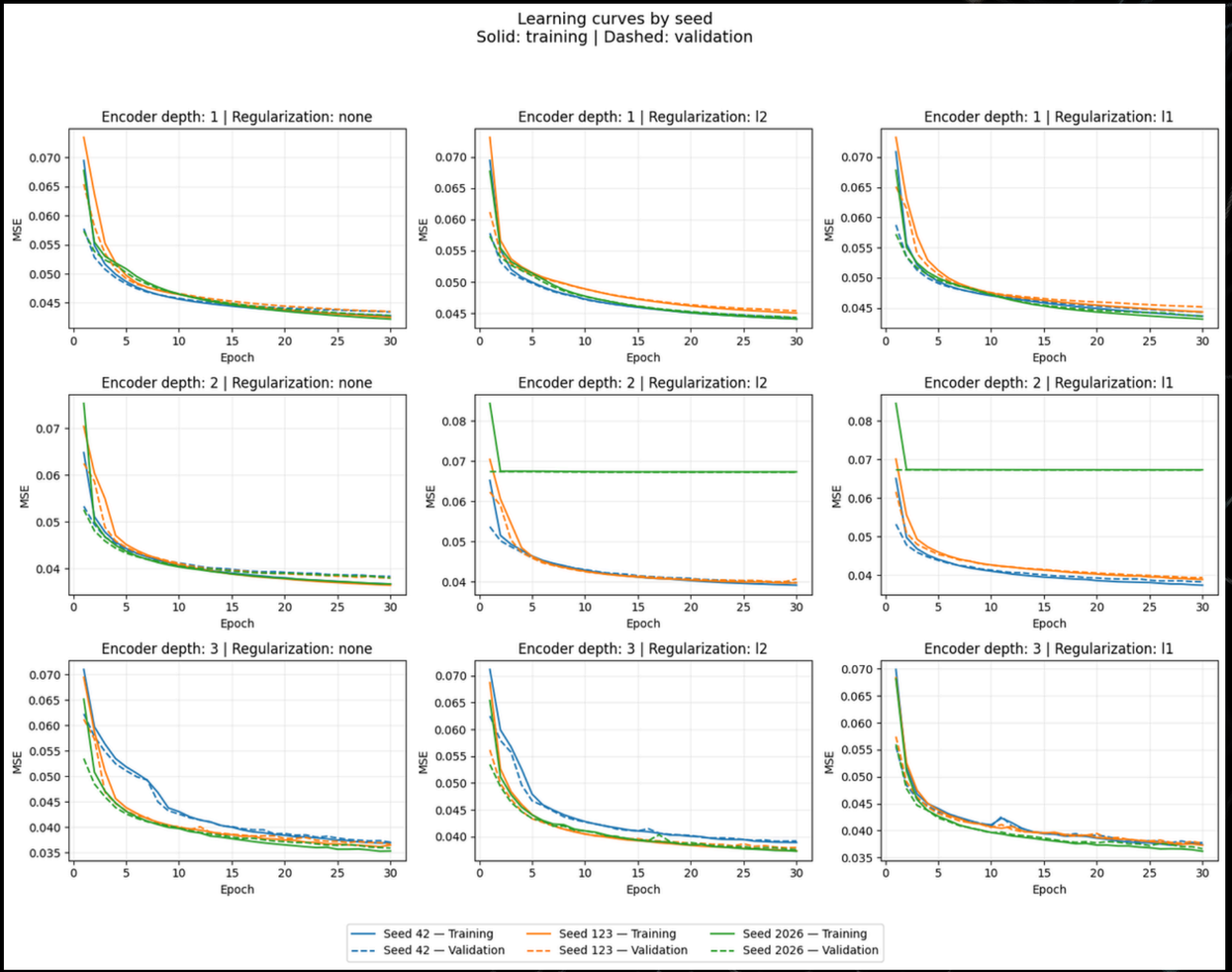


Figure 3 - Training and validation MSE across seeds.

Most runs improve steadily

Training and validation MSE decrease together, with relatively small gaps

Seed 2026 stalls in depth-2 regularized models

With L1 and L2, MSE remains near 0.067, while other seeds continue improving.

Multiple seeds reveal training instability

The stalled runs increase the mean validation MSE and its standard deviation

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TEST PERFORMANCE

Selected using validation performance

3 hidden layers with no regularization

Mean test MSE → 0.036351

Small variation across the three seeds

Similar validation and test performance

Validation MSE → 0.036497

Seed	Test MSE
42	0.036745
123	0.036413
2026	0.035895

Table 3 - Test MSE across seeds.

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TEST PERFORMANCE

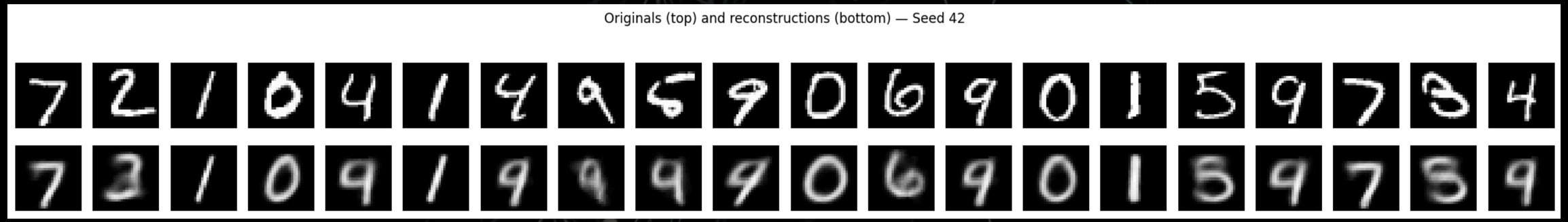


Figure 4 - Original test images (top) and reconstructions (bottom), using seed 42 of the selected configuration.

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LATENT SPACE & IMAGE SYNTHESIS

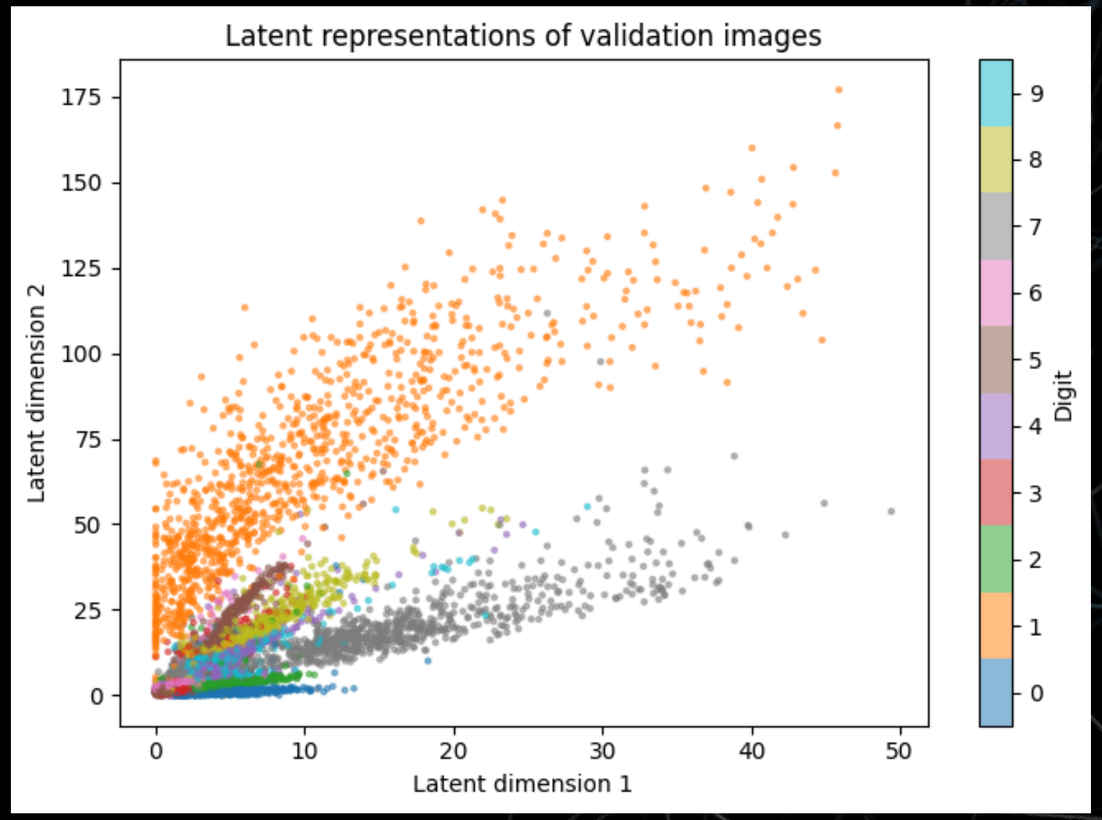


Figure 5 - Two-dimensional latent representations of validation images, colored by digit label. Selected configuration, seed 42.

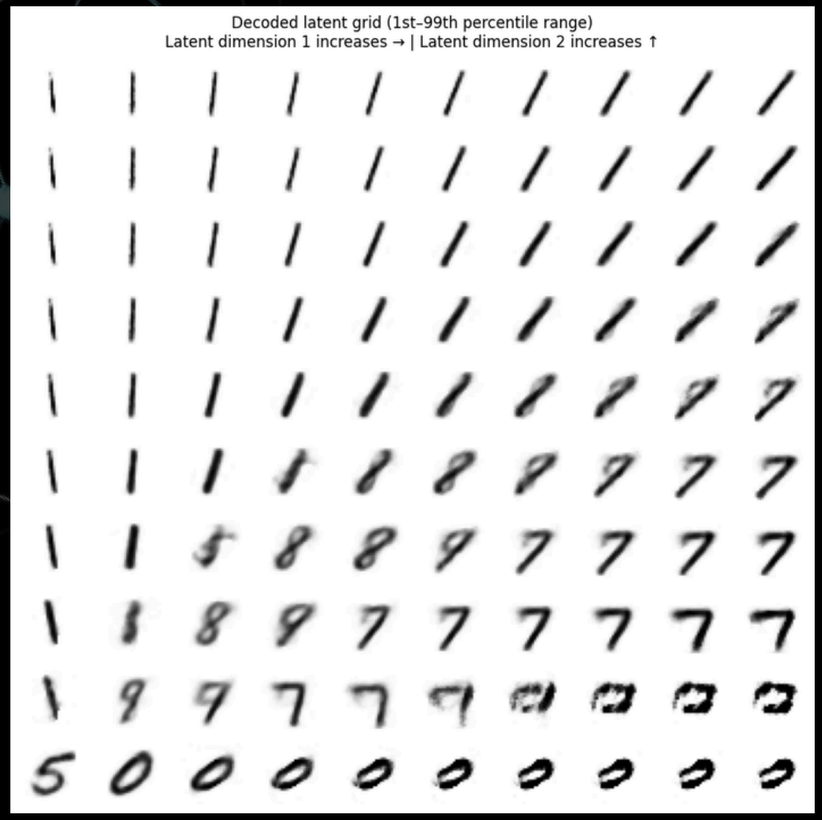
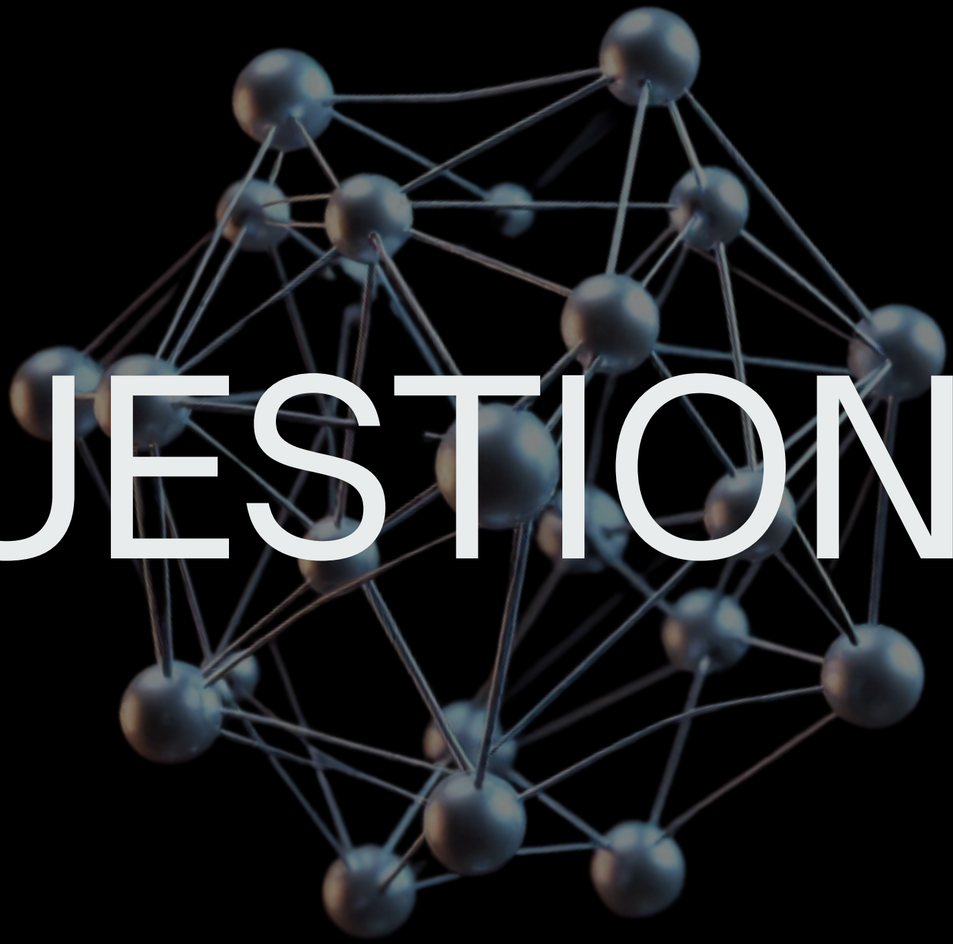


Figure 6 - Images decoded from a latent grid spanning the 1st–99th percentiles of each coordinate. Selected configuration, seed 42.



QUESTIONS?

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